

1. Administrative & Study Identification

Full Study Title: A Phase II, Open-label Study to Investigate the Pharmacokinetics and Safety of Risdiplam in Infants With Spinal Muscular Atrophy **Short Title/Acronym:** PUPFISH Study **Brief Title:** A Study to Investigate the Pharmacokinetics and Safety of Risdiplam in Infants With Spinal Muscular Atrophy **Protocol Number:** BN44619 (2023-505602-42-00) **NCT Number:** NCT05808764 **Posting ID:** 849699474298655348 **Sponsor/Company:** Roche (Hoffmann-La Roche) **Trial Status:** RECRUITING (Currently recruiting participants) **Phase of Development:** PHASE2 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) and safety of risdiplam in newborn infants with spinal muscular atrophy (SMA) who are under 20 days of age at their first dose. **Secondary Objectives:** To evaluate the tolerability, early efficacy, and pharmacodynamics (e.g., SMN protein expression levels) of risdiplam in this newborn population.

2.2 Background & Rationale Spinal muscular atrophy (SMA) is a severe, inherited neuromuscular disorder caused by a mutation in the survival motor neuron 1 (SMN1) gene, which leads to a deficiency of SMN protein, resulting in progressive muscle weakness and wasting. Risdiplam is a selective SMN2 pre-mRNA splicing modifier designed to increase and sustain the production of functional SMN protein. While risdiplam is approved for older infants and adults, the PUPFISH study is needed to specifically understand the drug's safety and PK profile in extremely young infants (under 20 days old) to optimize early intervention, which is critical for preserving motor neurons before significant irreversible damage occurs.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 10 **Number of Sites:** Approximately 13 locations globally (including sites in Canada, Netherlands, Belgium, Poland, and the US) **Estimated Study Start:** 26-Apr-2024 **Estimated Primary Completion:** 04-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female newborn infant aged < 20 days at first dose. 2. Newborn infants with genetic diagnosis of 5q-autosomal recessive SMA or newborn infants identified as positive for SMA via newborn screening or via prenatal testing. 3. Gestational age equal to or greater than 37 weeks. 4. Receiving adequate nutrition and hydration at the time of screening. 5. Adequately recovered from any acute illness at baseline and considered well enough to participate in the study. 6. Parent/caregiver is willing to consider nasogastric, nasojejunal, or gastrostomy tube placement during the study to maintain safe hydration, nutrition, and treatment delivery, if recommended by the investigator.

4.2 Exclusion Criteria 1. Presence of clinical symptoms or signs consistent with SMA Type 0. 2. In the opinion of the investigator, inadequate venous or capillary blood access for the study procedures. 3. Systolic blood pressure or diastolic blood pressure or heart rate abnormalities. 4. Presence of clinically relevant electrocardiogram (ECG) abnormalities. 5. The infant (or the person breastfeeding the infant) taking any of the following: any inhibitor of CYP3A4 taken within 2 weeks (or within 5 times the elimination half-life, whichever is longer) prior to dosing, any inducer of CYP3A4 taken within 4 weeks (or within 5 times the elimination half-life, whichever is longer) prior to dosing, and/or use of any multidrug and toxin extrusion (MATE) substrates taken within 2 weeks (or within 5 times the elimination half-life, whichever is longer) prior to dosing. 6. Concurrent or previous administration of nusinersen or onasemnogene abeparvovec. 7. Clinically significant abnormalities in laboratory test.

4.3 Protocol Criteria Requirements Criteria Type:

PHASE_REQUIREMENT Criteria Text: Trial must be in PHASE2

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Risdiplam **Trade Name:** Evrysdi **ATC Code:** M09AX10 **Dosage/Frequency:** Risdiplam administered once daily. **Route of Administration:** Oral (solution/dispersed). **Duration of Treatment:** 4 weeks of active treatment for the primary PK evaluation period. **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/evrysdi-epar-product-information_en.pdf

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for SMA including nutritional and respiratory support. **Prohibited:** Concurrent treatment with other disease-modifying SMA therapies (such as nusinersen or onasemnogene abeparvovec) during the evaluation period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Prior to Day 1	Informed Consent, Medical History, Genetic Testing/Verification, Eligibility Assessment
Baseline	Day 1	First Dose of risdiplam, Safety assessments, PK sampling
Interim	Days 2, 5, and 10	PK sampling, Safety Labs, Clinical assessments
End of Study	Day 28	Final PK sampling, Safety Follow-up, Transition to standard of care/extension

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Plasma Concentrations of Risdiplam. 2. Area Under the Plasma Concentration-Time Curve (AUC) of Risdiplam. 3. Steady-state Concentration (C_{ss}) of Risdiplam.

7.2 Secondary & Exploratory Endpoints 1. Incidence, severity, and relatedness of Adverse Events (AEs) and Serious Adverse Events (SAEs). 2. Change from baseline in SMN protein levels in blood (Pharmacodynamics).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via caregiver reports, physical exams, and clinical safety labs during site visits on Days 1, 2, 5, 10, and 28. **Serious Adverse Events (SAEs):** Typical 24-hour reporting timeline to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Internal/External DMC oversees safety data given the vulnerable newborn population.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Population (all participants receiving ≥ 1 dose with evaluable PK data) and Safety Set (all participants who receive at least one dose of risdiplam). **Sample Size Justification:** Target N=10 is considered adequate for exploring the PK/PD and safety profile in this rare disease, ultra-young age cohort (< 20 days old). **Handling of Missing Data:** Standard handling rules for missing PK time points (e.g., actual sampling times used for non-compartmental analysis).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Frequent onsite monitoring to ensure protocol

adherence, accurate PK sampling times, and safety verification.
Audit Trail: Standard tracking of eCRFs with data clarification forms.

11. Study Identifiers & Governance

Primary ID: BN44619 **Registry Number:** NCT05808764 **Posting ID:** 849699474298655348 **Source Level:** 5 **Sponsor/Lead Organization:** Roche **Study Title:** PUPFISH **Official Regulatory Title:** A Phase II, Open-label Study to Investigate the Pharmacokinetics and Safety of Risdiplam in Infants With Spinal Muscular Atrophy

12. Clinical Framework (Disease Specifics)

Disease Areas: Muscular Atrophy, Spinal **Preferred UMLS Name:** Spinal Muscular Atrophy **Semantic Type:** Pathologic Function **MeSH Code:** D009133 **SNOMED ID:** 88092000 **Medical Methodology:** SMN2 pre-mRNA splicing modifier **Optimization Target:** Systemic increase of functional survival motor neuron (SMN) protein

13. Study Procedures & Methodology

Management Pathway: Early intervention targeted therapy via daily oral administration **Education Protocol (HCP):** Site training on neonatal PK blood draw procedures and dosing administration. **Education Protocol (Patient/Caregiver):** Caregiver instruction on oral administration of risdiplam, monitoring for adverse events, and recording dosing times. **Quality Audit Mechanism:** Clinical monitoring and caregiver dosing diary review.

14. Observation & Follow-Up Schedule

Total Duration: 4 weeks of intensive active PK evaluation (28 days) **Onsite Visit Cadence:** Days 1, 2, 5, 10, and 28 **Remote Monitoring Frequency:** Continuous caregiver contact between visits if needed. **Checklist Review Interval:** At every scheduled study visit to ensure medication compliance.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				